

Study Guide: Disruptive Technology and Innovation Management

This study guide provides a comprehensive review of the provided research on disruptive technology business models, specifically focusing on cloud computing and the foundational theories established in Clayton M. Christensen's *The Innovator's Dilemma*.

Part 1: Short-Answer Quiz

Instructions: Answer the following questions in two to three sentences based on the provided text.

1. How is cloud computing defined within the context of disruptive business models?
 2. What is the fundamental difference between sustaining and disruptive technologies?
 3. According to the resource dependence theory, why do high-performing companies often fail to invest in disruptive innovations?
 4. Describe the "technology mudslide hypothesis" and why the author ultimately found it to be an incorrect explanation for firm failure.
 5. What are the three criteria used to characterize cloud computing as a disruptive technology?
 6. Why is it often a failing strategy for large companies to wait until a new market is "large enough to be interesting"?
 7. How does "discovery-based planning" differ from traditional market research and planning?
 8. In the context of organizational capabilities, how can a firm's values act as a "disability"?
 9. Explain the concept of "performance overshooting" in a mainstream market.
 10. How did the mechanical excavator industry illustrate the pattern of disruptive innovation during the transition from cable to hydraulic systems?
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Part 2: Answer Key

1. **Cloud Computing Definition:** It is defined as a scalable, flexible, shared computing solution where third-party suppliers use virtualization to distribute resources on-demand

via an Internet browser. It acts as a disruptive force by replacing rigid software licensing models with utility-based pricing and greater scalability.

2. **Sustaining vs. Disruptive:** Sustaining technologies improve the performance of established products along dimensions that mainstream customers already value. Disruptive technologies initially underperform in mainstream metrics but offer new attributes—such as being cheaper, smaller, or more convenient—that appeal to fringe or new customer segments.
3. **Resource Dependence:** This theory suggests that customers and investors effectively dictate how a company spends its money, as firms that do not satisfy them cannot survive. Consequently, well-managed companies find it difficult to allocate resources to low-margin, disruptive opportunities that their best customers do not currently want.
4. **Technology Mudslide Hypothesis:** This hypothesis suggested that firms fail because they cannot keep up with the rapid pace of technological change, similar to someone struggling to climb a mudslide. The research proved this wrong by showing that established firms actually lead in the most difficult sustaining technological changes, only stumbling when faced with simpler, disruptive ones.
5. **Three Criteria for Disruption:** First, the innovation must enable less-skilled or less-wealthy individuals to achieve the same utility previously reserved for experts or the wealthy. Second, it must target the low end of a market with a performance trajectory capable of eventually taking over higher tiers. Third, it requires an ecosystem—either a single entity or modular entities—to support the innovation.
6. **The Growth Needs Paradox:** To maintain stock prices and internal opportunities, large companies require significant revenue growth that small, emerging markets cannot initially provide. While a \$40 million company can thrive on \$8 million in new revenue, a \$4 billion company needs \$800 million, making emerging markets appear too insignificant to justify investment until it is often too late to gain a first-mover advantage.
7. **Discovery-Based Planning:** Traditional planning assumes that market sizes and returns can be quantified in advance, which is impossible for markets that do not yet exist. Discovery-based planning operates on the assumption that initial forecasts are likely wrong, prioritizing a process of learning and adapting the strategy as the market reveals itself.
8. **Values as Disabilities:** Organizational values are the criteria used by employees to prioritize projects; for example, if a firm's values are tuned to prioritize high-margin products for large customers, they automatically disable the organization from effectively pursuing low-margin, small-market opportunities. Processes and values that are successful in one context can become rigid barriers in a disruptive context.
9. **Performance Overshooting:** This occurs when the pace of technological progress exceeds the rate of performance improvement that mainstream customers can actually absorb. Eventually, the product becomes more advanced (and expensive) than what the user needs, creating a "vacuum" at the lower end of the market for simpler disruptive technologies to enter.
10. **Excavator Industry Example:** Leading manufacturers of cable-actuated shovels were successful at every sustaining innovation (like gasoline or diesel power) because their customers needed more power. However, they were nearly all wiped out by hydraulic

technology, which initially had very low capacity and could only be used for small residential trenches—a market the large manufacturers ignored.

Part 3: Essay Questions

Instructions: Use the provided source context to develop comprehensive responses to the following prompts.

1. **The Disk Drive Saga:** Analyze the historical transitions in the disk drive industry (from 14-inch to 1.8-inch drives). Explain how the "innovator's dilemma" manifested in these transitions and why "staying close to the customer" proved fatal for industry leaders.
 2. **The Mechanics of Failure:** Compare and contrast the failure of established firms in the computer industry with those in the mechanical excavator or integrated steel industries. Identify the common patterns in management decision-making that led to their displacement.
 3. **The Value Network Framework:** Define the concept of a "Value Network." Discuss how a firm's position within a specific network shapes its cost structure, its perception of "profitable" innovation, and its ultimate flexibility to respond to threats from other networks.
 4. **Strategic Management of Disruption:** Based on the principles outlined in the text, describe the organizational structures and planning methodologies a large company should employ to successfully commercialize a disruptive technology like the electric vehicle or flash memory.
 5. **Cloud Computing as a Case Study:** Using Christensen's principles, evaluate cloud computing as a disruptive technology. Discuss the "subordinate disruptive forces" mentioned in the text (such as hybrid clouds and vendor-agnostic solutions) and their role in integrating the cloud landscape.
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Part 4: Glossary of Key Terms

Term	Definition
Cloud Computing	A scalable, shared computing solution using virtualization to provide resources on-demand via a browser, typically using a utility-based pricing model.

Disruptive Technology	An innovation that brings a different value proposition to the market, typically underperforming in mainstream metrics but offering new features like low cost or simplicity.
Sustaining Technology	Innovations that improve the performance of established products along dimensions historically valued by mainstream customers.
Value Network	The context within which a firm identifies and responds to customer needs, solves problems, and reacts to competitors; it defines the firm's cost structure and metrics of value.
Performance Trajectory	The rate at which a product's performance improves over time within a specific technological paradigm or market application.
Resource Dependence	The theory that a company's freedom of action is limited by the requirements of the customers and investors who provide the necessary resources for survival.
Virtualization	A key technological capability in cloud computing used by third-party suppliers to create and distribute computing resources.
Discovery-Based Planning	A management approach for emerging markets that assumes forecasts are wrong and focuses on a process of learning and strategic adjustment.
Overshooting	The phenomenon where technological progress provides more performance than mainstream customers are able to absorb or willing to pay for.

Flash Memory	A solid-state semiconductor memory technology that retains data without power; used as a disruptive example that initially found success in non-computing markets.
Winchester Architecture	A fundamental disk drive design introduced by IBM in 1973 that became a dominant standard in the industry.
OEM (Original Equipment Manufacturer)	A market where disk drive makers sold their products to computer manufacturers rather than directly to end-users.